SAMAY THAKUR

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EDUCATION

M.S. Ramaiah Institute of Technology — Oct 2023 – Present
Bachelor of Engineering (B.E.) | Electronics and Instrumentation Engineering | CGPA: 7.95 — Bengaluru, India

Sunbeam School Varuna — Mar 2019 – Jun 2022
Grade X (CBSE): 89.6% | Grade XII (CBSE): 74.2% — Varanasi, India

EXPERIENCE

CPS & IoT Intern — Aug 2024 – Aug 2024
Intra Institutional Internship, MSRIT — Bengaluru, India

- Designed a real-time water-level monitoring and automated pump-control system using Arduino Uno and an ultrasonic sensor for non-contact water-level measurement.
- Implemented C++ threshold-based control logic to automate pump ON/OFF operations, helping prevent tank overflow and dry-running conditions.
- Validated system behavior across different water levels and edge cases to verify reliable sensor readings and automated pump control.

PROJECTS

SmartQC: On-Device Visual Quality Inspection | GitHub
Qualcomm | Snapdragon® AI Lab Build & Present Challenge

- Developed a PyQt5 desktop application integrating image classification, Grad-CAM explainability, batch inspection with CSV export, and SQLite-based inspection history with trend tracking.
- Fine-tuned MobileNetV2 for fresh-vs-mold bread classification, achieving 99.2% validation accuracy (0.9997 ROC-AUC), and optimized the model for on-device Snapdragon NPU deployment via Qualcomm AI Hub.
- Evaluated the classifier on 352 uncurated real-world images, achieving 97.4% field accuracy, and implemented a confidence-threshold mechanism to flag low-confidence predictions for manual review.
- Independently defined the problem statement, target users, success criteria, and evaluation approach, taking the project from initial scoping through implementation and evaluation.
- Tools/Technologies: Python, PyTorch, ONNX, Qualcomm AI Hub, PyQt5, OpenCV, scikit-learn, SQLite

Stock Market Price Prediction & Analysis | GitHub

- Developed an end-to-end Python pipeline to process 2019–2024 NIFTY 50 OHLCV data, engineer technical indicators, train an LSTM model, and evaluate next-day closing-price predictions.
- Engineered MA20/MA50, RSI, MACD, and Bollinger Bands features, conducting stationarity testing, correlation analysis, and sector-performance analysis to examine market patterns.
- Trained a two-layer TensorFlow/Keras LSTM using 60-day sliding windows and a chronological 80/20 train-test split, achieving 1.62% MAPE and 0.91 R^2 on the held-out test set.
- Independently managed the project lifecycle, from data preparation and model development to evaluation, documentation, and GitHub delivery; identified approximately 52% directional accuracy, highlighting limitations in practical forecasting performance.
- Tools/Technologies: Python, TensorFlow/Keras, LSTM, Pandas, NumPy, Scikit-learn, yfinance

CynLr EPM PoC | GitHub

- Structured an open-ended robotics PoC involving moving-object tracking and track-and-grasp into a program plan spanning robotics, CV/ML, software, mechanical, electronics, procurement, and external vendors.
- Mapped the execution plan in Asana across 9 workstreams, 41 activities, and 97 subtasks, defining task ownership, dependencies, milestones, risks, and Phase 1 Go/No-Go/Revise decision gates.
- Estimated 604 person-hours across 9 activities and 31 subactivities; defined 21 lead and lag KPIs to track dependency resolution, development progress, infrastructure readiness, testing, and delivery.
- Designed a progress-monitoring and escalation framework covering milestone reviews, dependency resolution, effort reassessment, and resource reallocation to assess quality and integration readiness.
- Tools/Technologies: Asana, Microsoft Excel

SKILLS

- **Programming Languages:** C++, Python
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks
- **Software Development Tools:** Git, GitHub, LabVIEW, VS Code, Jupyter Notebook
- **Frameworks & Libraries:** Flask, PyQt5, OpenCV, Pandas, NumPy, Scikit-learn
- **AI/ML & Model Deployment:** PyTorch, TensorFlow/Keras, ONNX, Qualcomm AI Hub, Computer Vision, Deep Learning (CNN, LSTM), Explainable AI (Grad-CAM)
- **Data & Analytics:** SQL, Excel, Tableau, Power BI, Exploratory Data Analysis, Time-Series Forecasting, Model Evaluation
- **Program Management & Leadership:** Asana, Program Planning, Cross-Functional Planning, Dependency Management, Risk Management, Milestone Tracking, KPI Tracking, Resource Planning

ACHIEVEMENTS & LEADERSHIP

- **Student Member:** Instrument Society of India (ISOI), IISc Bangalore (Nov 2023 – Oct 2027)
- **Community Volunteer:** Organized and supported blood-donation drives through school and community initiatives, facilitating donor engagement and community welfare.